

Homework 12 due Dec 3 before class

1. (10 pt). Refractive index of air is $n \approx 1.0003$. With what Lorentz factor a particle should have to produce Cherenkov emission?
2. (20 pt). Bending of light in inhomogeneous dielectric. Consider weakly inhomogeneous dielectric with $\epsilon = \epsilon^{(0)}(1 + \alpha z)$ ($\epsilon^{(0)}$ is normalization of ϵ , not the ϵ_0 of the SI). Weakly inhomogeneous means α is small, or in other words, the scale of change of ϵ $L \sim 1/\alpha$ is much larger than the wave length c/ω .

An electromagnetic with frequency ω propagates in a dielectric. The wave has a well-defined projection of the wave vector on perpendicular direction x , κ . Question: depending on α how does the light propagates: does it bend towards z axis or away?

Hint: The problem follows on Landau & Lifshitz, vol. VIII, prag. 88; Let's concentrate on the X-mode. Eq. 88.3. In the weakly inhomogeneous case,

$$\begin{aligned} E, B &\propto e^{i\Theta} \\ \mathbf{k} &= \nabla\Theta \end{aligned} \tag{1}$$

In the final expression do expansion $\alpha \rightarrow 0$.